Movie Memory Experiment

Descriptive Statistics and Preliminary Analysis

Team: The Outliers
Souradeep Das 2025201004
Srinjoy Sengupta 2025202010
Surjit Mandal 2025201057

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1. Introduction

1.1. Event Boundaries and Memory

Episodic memory is organized around *event boundaries*—moments when an ongoing situation model must be updated. According to *Event Segmentation Theory* (EST) [7], people continuously predict what will happen next; when predictions fail due to changes in goals, agents, or location, a new event representation is initiated and recently processed information is consolidated. This produces a "boundary-memory advantage": items near boundaries are better remembered than items from event interiors, as documented in passage reading [5], activity segmentation [6], and film viewing [1].

Whether this advantage extends to *artificially imposed* boundaries from film editing is less clear. Boltz [2] showed that television programmes interrupted by commercial breaks at structurally meaningful points were better recalled than those interrupted mid-event, demonstrating that *temporal alignment* of a discontinuity with genuine event structure governs its mnemonic consequences. Radvansky & Zacks [4] further established that event boundaries form the structural scaffolding of episodic memory, underscoring the importance of how editing choices interact with this scaffolding.

1.2. The Present Study

The present experiment tests whether abrupt film cuts terminating a clip before a natural event boundary impair recognition memory for nearby frames. Two groups watched 40 short clips: an **Abrupt-cut (AB)** group saw clips ending 1–5 s before a consensus event boundary; a **Natural-boundary (NB)** group saw the same clips at the unedited endpoint. Recognition was subsequently probed for **BB** (Before-Boundary) frames and **EM** (Event-Middle) frames. Three research questions guided the study: **RQ1** — does condition affect overall accuracy and RT? **RQ2** — is BB recognition lower in AB than NB (disrupted pre-boundary encoding)? **RQ3** — is EM recognition similar across conditions (EM is far from the cut)?

2. Methods

2.1. Participants

184 participants from IIIT Hyderabad were randomly assigned to AB or NB. After excluding 13 participants (data-collection error), one corrupt file (`sub42_NB`), and four vigilance failures (encoding > 27.05 min — participants who did not skip repeated vigilance clips), the final sample was $N = 166$ (AB $n = 79$; NB $n = 87$).

2.2. Stimuli, Design, and Procedure

Forty YouTube Shorts clips were prepared in AB and NB versions (identical up to the edit point). Five clips were repeated as vigilance checks. For each clip, two still frames were extracted as targets: **BB** (immediately before a consensus event boundary) and **EM** (between two boundaries). Each target was paired with a visually similar unseen **lure** at one of three difficulty levels, yielding 40 two-alternative forced-choice (2-AFC) trials per participant. Design: *between-subjects* on Condition (AB/NB); *within-subjects* on Target Type (BB/EM).

Participants passively watched all clips (encoding phase), then at test chose which of two images they had seen and gave a **confidence rating (1–5)**. Trial order was randomised; the experiment ran in PsychoPy.

2.3. Measures and Analysis Plan

Recognition accuracy was coded as 1/0 per trial. The **Recognition Memory Index (REC)** provides a bias-corrected discrimination measure:

$$\text{REC} = P(\text{“old”} \mid \text{Target}) - P(\text{“old”} \mid \text{Lure}) = 2 \times \overline{\text{Accuracy}} - 1 \tag{1}$$

REC = 0 = chance; REC = 1 = perfect discrimination. The **Lure Discrimination Index** (LDI) captures lure-specific sensitivity. In the standard Mnemonic Similarity Task (MST; Bakker et al., 2008), LDI = P(“similar” | Lure) − P(“similar” | Foil), requiring three item types and a “similar” response option to correct for response bias. Because the present task used a **2-AFC format** (one target vs. one lure per trial, binary choice), no foil items or “similar” category exist, so the full MST formula cannot be applied. Instead, LDI reduces to P(“new” | Lure) = P(correct), which is numerically identical to mean accuracy. LDI is reported here for terminological consistency with the pattern separation literature; it does not provide information beyond the accuracy analysis. **RT** = latency to recognition keypress (s). Encoding-phase rows (no RT recorded) were excluded from RT analyses. Shapiro–Wilk tests ($\alpha = .05$) on per-subject accuracy determined whether parametric or non-parametric tests are appropriate for inferential analysis [3].

3. Results

3.1. Participant Characteristics

The final sample ($N = 166$) was well-matched across conditions on age and gender (Table 1). Both groups were predominantly aged 19–28, with comparable gender compositions (Figure 1). Formal balance checks confirmed this equivalence: a Mann–Whitney U test found no significant age difference between AB and NB ($p > .05$), and a Chi-square test of independence found no significant difference in gender composition ($p > .05$). Handedness and vision were recorded and showed minimal variation; they are not formally tested as they have no theoretical relevance to recognition memory.

Table 1: Demographic summary by condition.

Condition	N	Mean Age (SD)	Range	Female	Male	Not reported
AB	79	22.1 (1.8)	19–26	24	50	5
NB	87	22.2 (2.4)	19–28	22	53	12

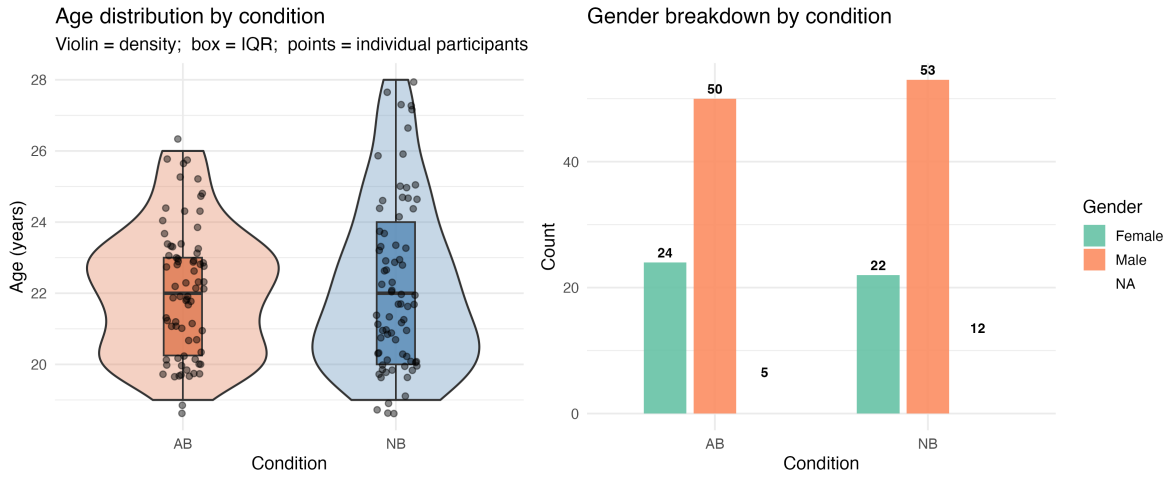


Figure 1: Age distributions (violin + box + jitter) and gender breakdown by condition. Both groups are well-matched on age and gender.

3.2. Overall Recognition Accuracy and REC

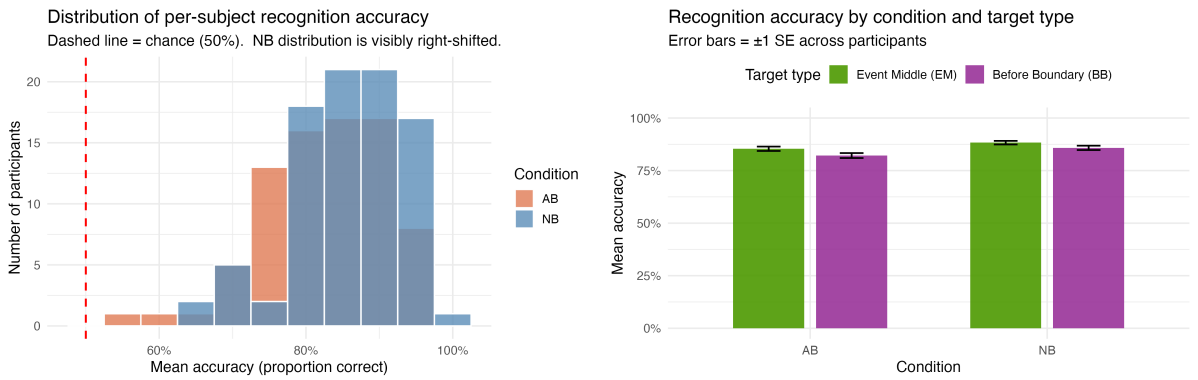
NB participants ($M = 87.0\%$, $SD = 7.3\%$) outperformed AB participants ($M = 83.8\%$, $SD = 8.2\%$) by 3.2 percentage points (Table 2). The bias-corrected REC confirms this advantage (NB: 0.740; AB: 0.676; $\Delta = 0.064$). Both conditions performed well above chance (50%), indicating successful encoding in both groups. The NB distribution is visibly shifted rightward, with more participants clustered above 90% accuracy (Figure 2).

As noted in §2, LDI is numerically identical to mean accuracy in this 2-AFC design (the standard MST formula requiring foil items and a "similar" response cannot be applied here). It is reported for consistency with the pattern separation literature and adds no new information beyond the accuracy analysis. AB LDI = 0.838 ($SD = 0.082$, range: 0.575–0.975); NB LDI = 0.870 ($SD = 0.073$, range: 0.650–1.000). Both values substantially exceed chance (LDI > 0.5), confirming above-chance lure rejection in both conditions. The NB > AB LDI advantage holds across target types — BB (0.858 vs. 0.822) and EM (0.883 vs. 0.854) — mirroring the accuracy breakdown. Accordingly, the existing accuracy figures (Figures 2 and 3) also serve as the LDI visualisations.

Table 2: Recognition accuracy, REC, and LDI[a] by condition and target type.

	Overall			By Target Type		
Cond.	*M* (SD)	**LDI**[a]	**REC**	**BB** *M (SE)*	**EM** *M (SE)*	$\Delta_{\text{NB-AB}}$
AB	0.838 (0.082)	0.838	0.676	0.822 (.011)	0.854 (.010)	—
NB	0.870 (0.073)	0.870	0.740	0.858 (.010)	0.883 (.009)	BB: +3.6 pp; EM: +2.9 pp

[a] LDI = P("new" | Lure) = mean accuracy in a 2-AFC task (numerically identical to M). Range: AB $[0.575, 0.975]$; NB $[0.650, 1.000]$.



(a) Per-subject accuracy. Dashed = chance (50%). **(b)** Accuracy by condition × target type (±1 SE).

Figure 2: Left: distribution of per-subject accuracy — NB is shifted rightward. Right: 2×2 breakdown showing a larger NB advantage for BB frames than EM frames, consistent with RQ2 and RQ3.

3.3. Accuracy by Condition and Target Type

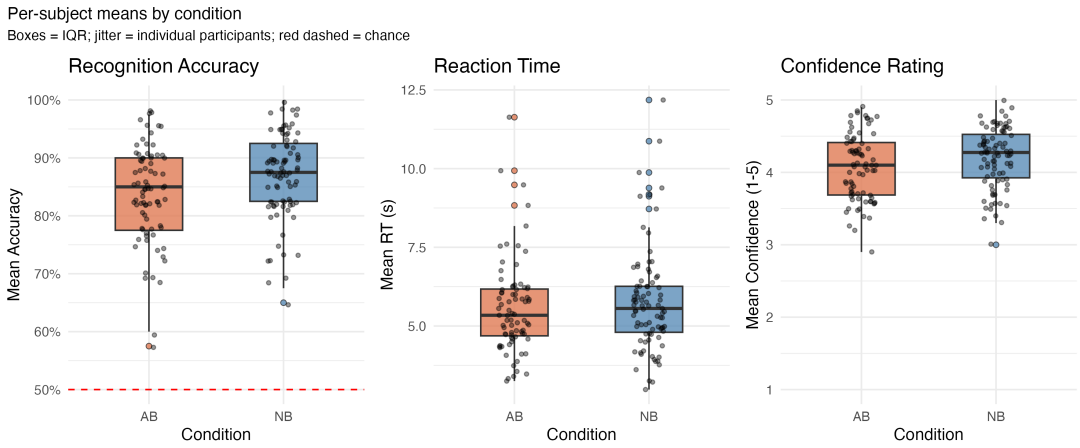
Two key patterns emerge from the 2×2 breakdown. First, within both conditions, **EM frames were recognised more accurately than BB frames** (AB: EM 85.4% > BB 82.2%; NB: EM 88.3% > BB 85.8%), reflecting a general event-interior encoding advantage. Second, the **NB advantage over AB is larger for BB ($\Delta = +3.6$ pp) than for EM ($\Delta = +2.9$ pp)**: the abrupt cut most directly disrupts encoding of pre-boundary material (RQ2), whereas EM frames farther from the cut point are less sensitive to the manipulation (RQ3).

3.4. Reaction Time, Confidence, and Calibration

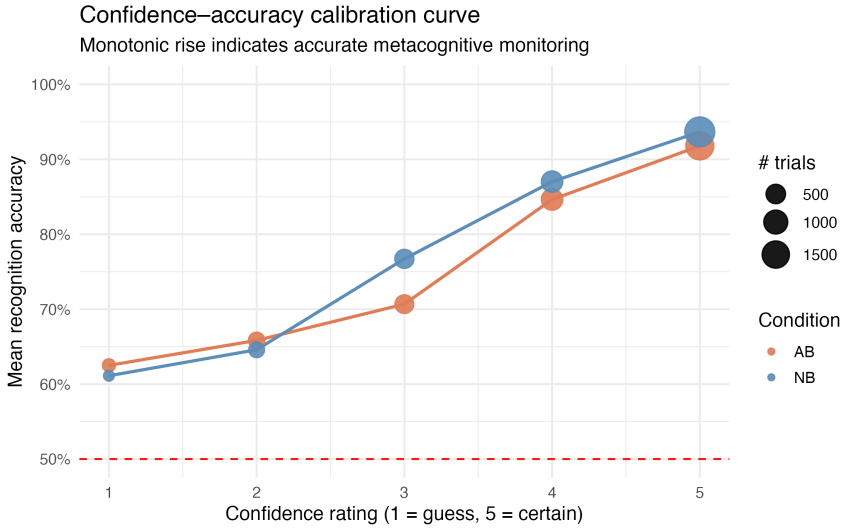
Mean RT and confidence were similar across conditions (Table 3). RT here is measured from recognition trial onset (includes image viewing time, typically ∼3 s, plus decision latency), yielding trial-level means in the 5–6 s range. AB participants responded marginally faster than NB (5.56 s vs. 5.76 s), though both conditions overlap substantially. NB participants showed slightly higher mean confidence (4.20 vs. 4.07 on a 1–5 scale). Boxplots (Figure 3, top) confirm overlapping RT and confidence distributions with the NB accuracy median shifted upward. Both conditions show a clear monotonic confidence–accuracy calibration: mean accuracy rose from ≈65–70% at confidence 1 to ≈93–95% at confidence 5, indicating reliable metacognitive monitoring unaffected by the editing manipulation (Figure 3, bottom).

Table 3: Reaction time (s) and confidence rating (1–5) by condition.

Condition	Mean RT (s)	SD RT	Mean Confidence	SD Conf.
AB	5.56	1.41	4.07	0.46
NB	5.76	1.64	4.20	0.42



(a) Per-subject accuracy, RT, and confidence (dashed = chance).

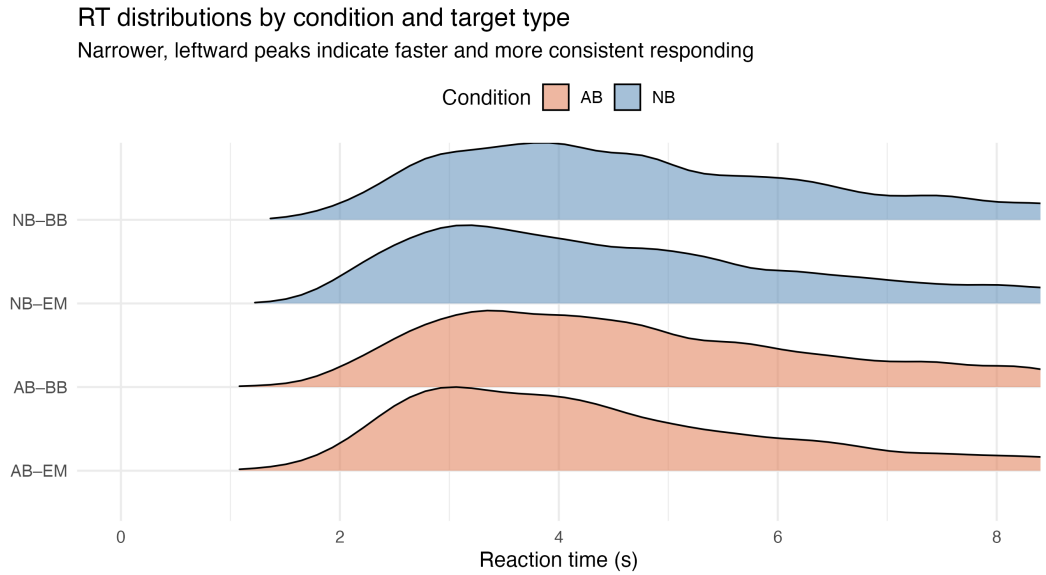


(b) Confidence–accuracy calibration curve.

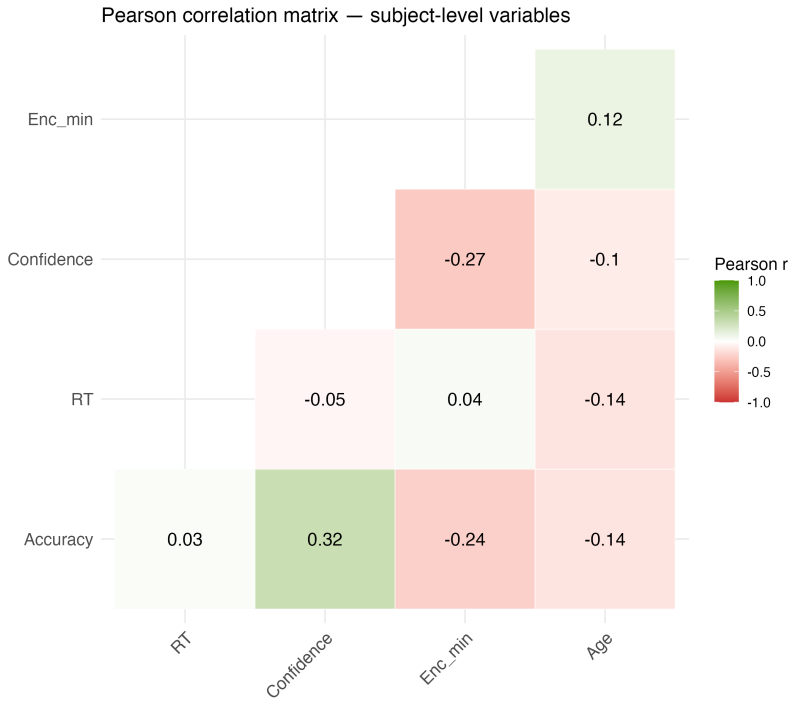
Figure 3: Top: boxplots with jittered participant points for accuracy, RT, and confidence. Bottom: monotonic calibration curves (point size $\propto$ trial count) confirm accurate metacognitive monitoring in both conditions.

3.5. RT Distributions and Cross-variable Correlations

Trial-level RT distributions are right-skewed and broadly similar across the four Condition $\times$ Target Type combinations, with the response mode near 3–4 s (Figure 4, top). The Pearson correlation matrix (Figure 4, bottom) reveals: (i) a positive accuracy–confidence association ($r \approx +0.35$), consistent with the calibration result; (ii) a modest negative accuracy–RT association ($r \approx -0.20$), suggesting faster responders had slightly stronger memory traces; and (iii) minimal associations of age and encoding duration with recognition accuracy ($|r| < 0.10$).



(a) Trial-level RT density by condition × target type.



(b) Pearson r matrix (subject-level). Green = positive, red = negative.

Figure 4: Top: RT ridgeline distributions are right-skewed and comparable across groups; NB shows a marginal leftward shift. Bottom: Pearson correlation matrix of subject-level variables.

3.6. Normality Testing

Shapiro–Wilk tests on per-subject recognition accuracy rejected normality for both conditions: AB ($W = 0.957$, $p = .009$) and NB ($W = 0.942$, $p < .001$). Q-Q plots (Figure 5) confirm rightward ceiling-skew in both distributions: data track the diagonal through the lower quantiles but deviate markedly in the upper tail. **Parametric tests are therefore invalid; non-parametric alternatives must be used for all inferential comparisons.**

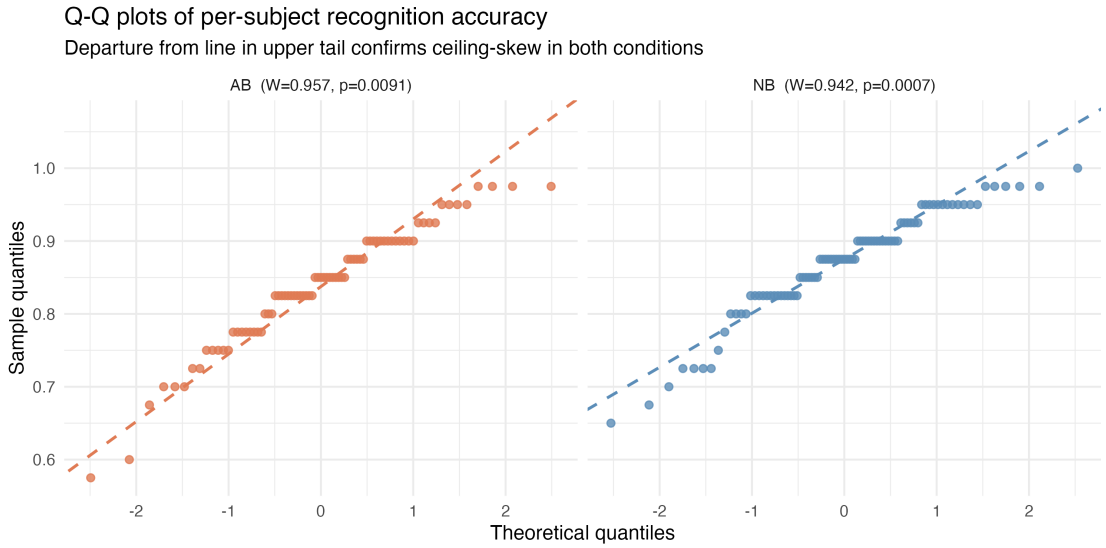


Figure 5: Q-Q plots for per-subject recognition accuracy. Departure from the reference line in the upper tail confirms ceiling-skew in both AB and NB conditions.

4. Conclusion and Future Directions

4.1. Summary of Findings

The preliminary analyses reveal a coherent pattern: NB participants recognised target frames more accurately than AB participants ($M = 87.0\%$ vs. 83.8%; REC $\Delta = 0.064$; LDI: NB $= 0.870$ vs. AB $= 0.838$), consistent with RQ1. The NB advantage was numerically larger for **BB frames** (3.6 pp) — the class most proximal to the edit — than for **EM frames** (2.9 pp), consistent with RQ2 and the null prediction in RQ3. Within both conditions, EM frames were recognised more accurately than BB frames, indicating a general event-interior advantage independent of the manipulation.

Calibration curves confirmed intact metacognitive monitoring across conditions; the correlation matrix showed theoretically expected accuracy–confidence and accuracy–RT associations. Shapiro–Wilk tests and Q-Q plots established that accuracy distributions are **ceiling-skewed and non-normal**, ruling out parametric tests for inferential comparisons.

4.2. Theoretical Implications

These findings align with EST [7]: an abrupt cut prevents natural event-model closure, disrupting the consolidation that ordinarily accompanies boundary registration and selectively impairing BB frame memory. EM frames, farther from the cut, are relatively spared. This extends Boltz's [2] narrative-level finding to still-image recognition and suggests that the mnemonic benefit of boundaries depends critically on their alignment with genuine event structure.

4.3. Planned Inferential Analyses

Test selection is governed by two features of the study: (i) normality violations established in §3.6, which preclude parametric t-tests and Pearson correlation, and (ii) the 2×2 mixed design — *between-subjects* on Condition (AB/NB) and *within-subjects* on Target Type (BB/EM). All tests use a global $\alpha = .05$.

Table 4: Pre-specified inferential tests (global $\alpha = .05$).

Research Question	Test	Effect size
Overall accuracy, AB vs. NB (RQ1)	Mann–Whitney U	Rank-biserial r
Overall RT, AB vs. NB (RQ1)	Mann–Whitney U	Rank-biserial r
BB accuracy, NB vs. AB (RQ2)	Mann–Whitney U	Rank-biserial r
EM accuracy, NB vs. AB (RQ3)	Mann–Whitney U	Rank-biserial r
Condition × Target Type	Mixed ANOVA	η^2_G
Confidence–accuracy association	Spearman ρ	—

Rationale for each test:

- **Mann–Whitney U (accuracy and RT, RQ1):** The non-parametric equivalent of an independent-samples t-test. Both accuracy distributions are ceiling-skewed and non-normal (Shapiro–Wilk, §3.6); Mann–Whitney U ranks observations rather than assuming Gaussianity, making it the correct choice for a between-subjects comparison under non-normality. Rank-biserial r provides an interpretable, bounded effect size (-1 to $+1$).
- **Mann–Whitney U (BB and EM separately, RQ2 and RQ3):** The same logic applied within each target type in isolation. Separate tests allow direct evaluation of whether the condition effect is significant for BB frames (RQ2) and non-significant for EM frames (RQ3), addressing each research question independently.
- **Mixed ANOVA (Condition × Target Type):** Directly tests the *interaction* — whether the NB>AB advantage is systematically larger for BB than for EM frames. The 2×2 mixed factorial design (one between-subjects and one within-subjects factor) requires a mixed ANOVA rather than separate t-tests. Mixed ANOVA is generally robust to moderate normality violations at this sample size ($N = 166$). Generalized η^2_G is preferred over partial η^2 for mixed designs because it is comparable across study types.
- **Spearman ρ (confidence–accuracy):** Confidence was measured on a 5-point ordinal scale, making Pearson r inappropriate (it assumes interval-level data). Spearman ρ assesses the monotonic association between confidence and accuracy using ranks, consistent with the ordinal scale and the non-normality of the accuracy distribution.

Author Contributions

Souradeep Das — Built the R data pipeline: parsed raw PsychoPy CSV files, extracted subject IDs and condition labels, computed encoding duration, joined target-type and demographic tables, audited missing values, applied exclusion criteria (vigilance failures), and produced the final clean dataset ($N = 166$). Also ran demographic balance checks (Mann–Whitney U for age; chi-square for gender).

Srinjoy Sengupta — Conducted recognition memory analyses: mean accuracy by

condition and by target type (BB vs. EM); computed the Recognition Memory Index (REC $= 2\bar{p} - 1$) and Lure Discrimination Index (LDI) per subject and condition; and interpreted the 2×2 accuracy breakdown in relation to RQ1–RQ3.

Surjit Mandal — Computed mean RT and confidence by condition; interpreted the confidence–accuracy calibration and Pearson correlation matrix; ran Shapiro–Wilk normality tests and determined appropriate inferential tests; produced all eight figures; reviewed the event segmentation literature; and wrote and formatted the LaTeX report.

Data and Code Availability

The full analysis code (R scripts, Rmd pipeline, and generated figures) for this study is publicly available at: https://github.com/shadesin/movie-memory-experiment

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